08 External Validation - eICU-CRD External Validation

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0.1 Introduction

External validation on a distinct dataset is the minimum standard for generalisability (Collins et al., 2024, TRIPOD+AI). The eICU-CRD (Pollard et al., 2018) provides multi-centre US ICU data with different demographics, protocols, and data infrastructure.

Expected AUC degradation: ~0.085 (Moor et al., 2023). This analysis tests **H5**: Utility Score degrades proportionally more than AUROC on external data.

0.1.1 Two anchor arms, and why both are needed

Sepsis-3 onset is anchored on the co-occurrence of an antibiotic order and a microbiology culture. In eICU-CRD that conjunction is only *recoverable* where microbiology exists, and `microLab.csv.gz` covers a small minority of unit stays. Applying the conjunction to the full cohort therefore does not measure a low sepsis incidence; it measures the coverage of one source table, because the overwhelming majority of person-hours sit in stays where the anchor cannot fire by construction. The two arms separate those questions:

Arm	Cohort	Anchor	Role
Sepsis-3 (primary)	stays with ≥ 1 culture record	culture $\wedge$ antibiotic, per-variant windows	label matches MIMIC-IV exactly; denominator contains only person-hours in which an onset *could* be observed
Antibiotic-only (sensitivity)	full cohort	first systemic antibiotic, no culture requirement	quantifies how much of the event count the culture requirement removes; the target is **not** Sepsis-3 and is never pooled with the primary arm

A third arm is reported in the final section of this notebook, separately from both of these. It applies the treatment-independent label (Variant D) to the full cohort. It is neither a Sepsis-3 estimate nor a sensitivity check on one: it is a different target, it is post-hoc under Protocol Amendment 9, and it is excluded from H5.

Steps:

1. Build the eICU person-hour panel harmonised to MIMIC-IV naming
2. Derive onsets **per label variant** under each anchor, using the same SOFA-delta criterion as `02_cohort_labels.Rmd`
3. Load MIMIC-IV trained models (no retraining)
4. Evaluate with the same metric suite
5. Reconcile the eICU and MIMIC-IV event rates explicitly

```
## Pre-registered expected degradation: 0.085 AUC
```

0.2 Load eICU Patient Table

Same inclusion filters as MIMIC-IV: age ≥ 18, LOS ≥ 4 hours.

```
## eICU patients (raw): 200,859
```

```
## eICU cohort after exclusions: 181,589
```

0.3 Vital Signs (vitalPeriodic)

Loaded via DuckDB with pushdown filters on cohort membership and time window.

```
## vitalPeriodic (cohort-filtered): 87,272,193 rows
```

```
##            used  (Mb) gc trigger   (Mb) limit (Mb)   max used    (Mb)
## Ncells  2510691 134.1    4392562  234.6         NA    4068927   217.4
## Vcells 69068380 527.0 1183558644 9029.9      16384 1476931673 11268.1
```

0.4 Lab Values

Lab values are mapped from eICU names to MIMIC-IV conventions. PaO_2/FiO_2 ratios computed where both components exist.

0.5 Drug Exposure: Antibiotics and Vasopressors

eICU records drugs across two tables. `medication.csv.gz` holds physician orders and `infusionDrug.csv.gz` holds continuous infusions; vasopressors and a large share of the intravenous antibiotics appear only in the latter. Both are read, and the antibiotic exposure is the union.

```
## Antibiotic orders (medication): 188,466 rows, 59,244 stays
```

```
##            used  (Mb) gc trigger   (Mb) limit (Mb)   max used    (Mb)
## Ncells  2519017 134.6    4392562  234.6         NA    4392562   234.6
## Vcells 97191549 741.6  605982027 4623.3      16384 1476931673 11268.1
```

```
## Antibiotic infusions (infusionDrug): 1,549 rows, 325 stays
## Vasopressor infusion hours: 465,746 (24,446 stays); norepi/epi: 344,314 (18,971 stays)
```

```
##            used  (Mb) gc trigger   (Mb) limit (Mb)   max used    (Mb)
## Ncells  2519259 134.6    4392562  234.6         NA    4392562   234.6
## Vcells 98816798 754.0  484785622 3698.7      16384 1476931673 11268.1
```

```
## Antibiotic exposure (union): 171,526 rows, 59,304 stays
```

0.6 Microbiology Coverage

`microLab.csv.gz` is the binding constraint on Sepsis-3 label transport. The coverage figure below defines the primary arm's cohort: a stay with no culture record cannot satisfy the Sepsis-3 conjunction under any variant, so counting its person-hours in the denominator measures table coverage rather than incidence.

```
## Culture records: 16,996 over 2,923 stays
```

```
## Microbiology coverage within the analysis cohort: 2,824 / 181,589 stays (1.56%)
```

```
## Culture records drawn before ICU admission (negative offset): 46.7%
```

0.7 Build eICU Person-Hour Panel

The panel is built **once**, from physiology alone, and carries no label. Onsets are derived from it per variant in the next section, which is what makes the three external label sets genuinely distinct. Construction mirrors MIMIC-IV (hourly grid, plausibility filter, LOCF, missingness indicators, rolling slopes, clinical scores). GCS is unavailable in `vitalPeriodic` and is left as `NA` rather than imputed.

```
##     hr               1253 values outside [20, 300] -> NA
##     resp_rate        2130 values outside [0, 80] -> NA
##     spo2             3853 values outside [50, 100] -> NA
##     map              4782 values outside [10, 250] -> NA
##     sbp              1862 values outside [30, 300] -> NA
##     fio2             3512 values outside [0.21, 1] -> NA
##     lactate          1 values outside [0, 40] -> NA
##     creatinine       17 values outside [0, 30] -> NA
##     bilirubin_total  1 values outside [0, 70] -> NA
```

Table 1: eICU onsets by label variant and anchor definition

variant	variant_name	anchor	n_stays	n_suspicion	n_onsets	pct_onsets	median_onset_h
A	Narrow	sepsis3	2824	186	27	0.96	14
A	Narrow	abx_only	181589	55273	5882	3.24	16
B	Seymour-Standard	sepsis3	2824	196	31	1.10	15
B	Seymour-Standard	abx_only	181589	55273	6608	3.64	16
C	Liberal	sepsis3	2824	211	73	2.58	9
C	Liberal	abx_only	181589	55273	9985	5.50	9

```
##     platelets        2 values outside [0, 2000] -> NA
##     wbc              7 values outside [0, 500] -> NA
##     pao2             39 values outside [20, 700] -> NA
##     pf_ratio         2318 values outside [10, 1000] -> NA
##     sodium           3 values outside [80, 200] -> NA
##     potassium        3 values outside [1, 12] -> NA
##   Plausibility filter (eICU panel): 19783 values set to NA
```

```
## eICU panel: 7,875,057 person-hours over 181,589 stays
```

0.8 Per-Variant Onset Derivation

Each label variant is applied to eICU with **its own** antibiotic–culture window and **its own** SOFA evaluation window and baseline, exactly as `02_cohort_labels.Rmd` applies them to MIMIC-IV. Two points of construction matter:

- The organ-dysfunction criterion is a SOFA **increase of ≥ 2 points over a baseline**, not an absolute score. An absolute threshold is a different criterion, and because `score_sofa()` substitutes assumed-normal values for missing components it also rejects on missingness rather than on physiology. The baseline is the admission window (hours 0–3) for Variants A and B and the rolling 24 hours preceding the evaluation window for Variant C, matching each variant's `sofa_baseline` setting.
- Suspicion drawn before ICU admission (a negative offset) makes the stay **non-septic within the panel**; it does not remove the stay. The same holds for suspicion after the stay ends. This mirrors the MIMIC-IV rule that an onset outside `[intime, outtime]` is set to `NA` while the stay remains in the at-risk denominator.

0.8.1 Variant-invariance guard

The failure mode this guards against is a label built once and reused across the variant loop: the external "variants" then differ only in which model is applied, every label-dependent quantity is identical across A, B and C, and the identity reads as a substantive finding rather than as the absence of a computation. A label-invariant comparator such as NEWS2 makes it visible — its AUROC can only be identical across variants if the label sets are identical — so the check is enforced at run time rather than left to inspection.

0.9 Apply MIMIC-IV Trained Models

Trained models are applied without retraining. Missing features are zero-filled (same as training). Predictions are generated in chunks for memory.

Table 2: eICU labelling-input coverage

n_cohort_stays	n_micro_covered_stays	pct_micro_covered	n_abx_stays	pct_abx_covered	n_both_
181589	2824	1.555	55273	30.439	

```
##   Feature check: all 22 declared features present.
##   Feature check: all 22 declared features present.
##   Feature check: all 22 declared features present.
##   Feature check: all 22 declared features present.
##   Feature check: all 22 declared features present.
##   Feature check: all 22 declared features present.
```

0.10 Results

Two arms are reported and they answer different questions. The primary arm applies the Sepsis-3 anchor to the sub-cohort where microbiology is documented. The sensitivity arm drops the culture requirement and runs on the full eICU cohort, which buys events at the price of no longer targeting Sepsis-3. Its rows bound what coverage costs; they are not an external Sepsis-3 result.

0.10.1 Event-rate reconciliation

An external event rate two or more orders of magnitude below the development cohort's is a property of the label extraction, not of the population. The table below states the rates side by side so the comparison is made explicitly rather than left to the reader.

0.10.2 H5: internal versus external AUROC

The internal-to-external AUROC drop per variant, set against the degradation Moor et al. give reason to expect. This is the AUROC limb standing in for H5 under Protocol Amendment 1, so what appears here is the descriptive contrast; the corrected verdict comes from stage 12. Where a variant's external event count falls under the interpretability floor the row is printed with a warning rather than suppressed, because the count is itself the finding.

```
##
## H5 test: AUROC internal -> external
##   VA: Internal AUROC=0.771, External AUROC=0.691, drop=0.080 (27 external events)
##   VA: Expected drop ~ 0.085 (Moor et al., 2023)
##   ** VA: 27 external events is below the 100-event floor; the estimate is not interpretable *
##   VB: Internal AUROC=0.759, External AUROC=0.696, drop=0.063 (29 external events)
##   VB: Expected drop ~ 0.085 (Moor et al., 2023)
##   ** VB: 29 external events is below the 100-event floor; the estimate is not interpretable *
##   VC: Internal AUROC=0.769, External AUROC=0.737, drop=0.032 (73 external events)
##   VC: Expected drop ~ 0.085 (Moor et al., 2023)
##   ** VC: 73 external events is below the 100-event floor; the estimate is not interpretable *
##
## External validation complete.
```

0.11 Treatment-Independent Label on the Full eICU Cohort (Variant D)

Everything above this point is confined to the microbiology-covered sub-cohort or, in the sensitivity arm, targets something other than Sepsis-3. Both limits have the same cause: the Sepsis-3 conjunction needs a culture record, and `microLab.csv.gz` reaches 1.56% of the analysis cohort.

Variant D (Protocol Amendment 2) does not have that dependency. Its onset is the first hour at which a **physiology-only** SOFA score stands 2 or more points above the stay's admission baseline, with no infection criterion and with the vasopressor and catecholamine columns withheld from `score_sofa()`. It contains no antibiotic order, no culture draw and no drug administration, so it can be constructed on **every** stay in the cohort rather than on the 2,824 where microbiology happens to be documented. That is the only reason this arm exists: it is the sole external estimate in the study whose denominator is the whole of eICU-CRD.

Three constraints on how the result may be read, enforced below rather than left to the reader:

- **Variant D is post-hoc.** It is outside the pre-registered A/B/C family, it is excluded from H1 and from the variance decomposition, and this arm is excluded from **H5** on exactly the same terms. It is written to its own output file, and the guards below assert that neither file has leaked into the other. It is descriptive: no p-value is attached and it joins no corrected family (Protocol Amendment 9).
- **The target is not Sepsis-3.** D measures acute organ dysfunction, not suspected-infection-with-organ-dysfunction. Its AUROC is never comparable with the primary arm's and the two are never pooled.
- **The two cohorts do not score SOFA from the same components.** eICU's `vitalPeriodic` carries no GCS, so the neurological component is scored from the assumed-normal substitution in both the hourly score and the baseline, where it cancels from the difference. eICU's deterioration delta therefore rests on five SOFA components and MIMIC-IV's on six. This is a genuine asymmetry between the development and validation cohorts, and it belongs beside the number.

The derivation mirrors `derive_deterioration_onsets()` in `03_person_hours.Rmd` — same physiology-only component set, same admission baseline window, same minimum observation floor — so that the external label is the same construct as the internal one rather than a second definition that happens to share a letter.

```
##   eICU Variant D: 52,535 / 181,589 stays deteriorate (28.93%), median onset hour 12.0
##   Feature check: all 22 declared features present.
```

0.11.1 Event rate under a treatment-independent label

The Sepsis-3 arm's external event rate is roughly a tenth of the development cohort's, and Section~**??** attributes that to the culture limb rather than to the population. Variant D tests the attribution directly: it is the same physiology in the same two databases with the treatment anchor removed. The rate is written to its own file so that the pre-registered reconciliation table is untouched.

0.11.2 Containment guards

Variant D is kept out of H5 and the variance decomposition the same way it is kept out of H1. The separation is structural — the confirmatory consumers read `08_external_validation_results`, and this arm writes somewhere else — but a structural argument is only as good as the next edit, so it is asserted at run time as well.

```
##
```

```
## Variant D external: 52,535 events on 181,589 stays (interpretability floor 100): above the fl
##
## Treatment-independent external arm complete.
```